SOC LAB PORTFOLIO-LAW-SOC-LAB

Project 1: Azure Sentinel SIEM Deployment & Initial Configuration

Objective:

Deploy a cloud-based Security Operations Center using Microsoft Azure Sentinel to simulate enterprise threat monitoring and incident response.

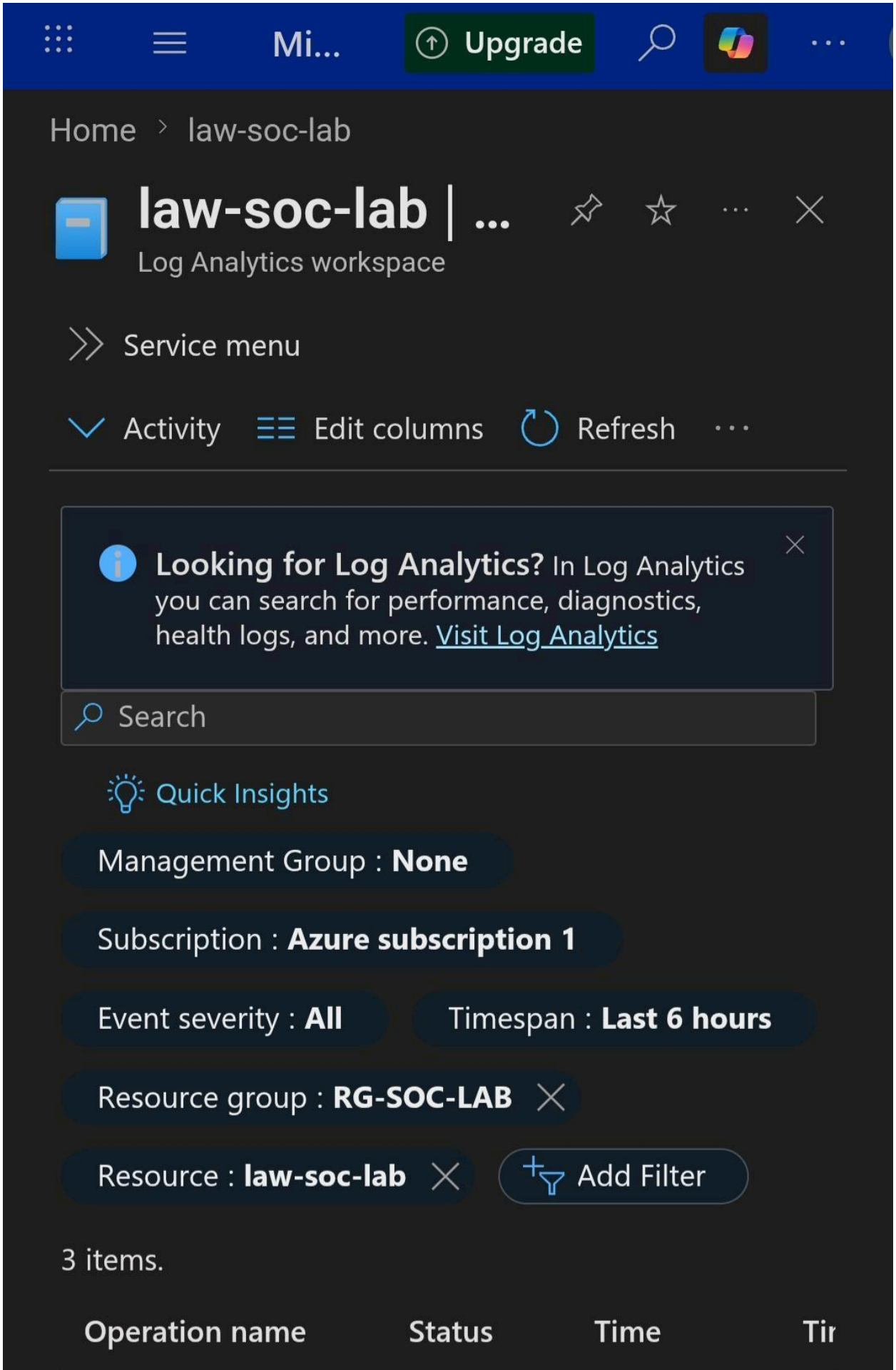
Environment Details:
Platform: Microsoft Azure
SIEM: Microsoft Sentinel
Workspace Name: law-soc-lab
Resource Group: SOC-Lab-RG
Region: South Africa North
Tasks Completed:

1Created Microsoft Azure Free Tier account with $200 credit
Established new Microsoft Entra ID tenant for lab isolation
Configured Global Administrator role for full control

2. Log Analytics Workspace Provisioning
Deployed Log Analytics Workspace named law-soc-lab
Configured data retention and pricing tier
Verified workspace health and data ingestion readiness

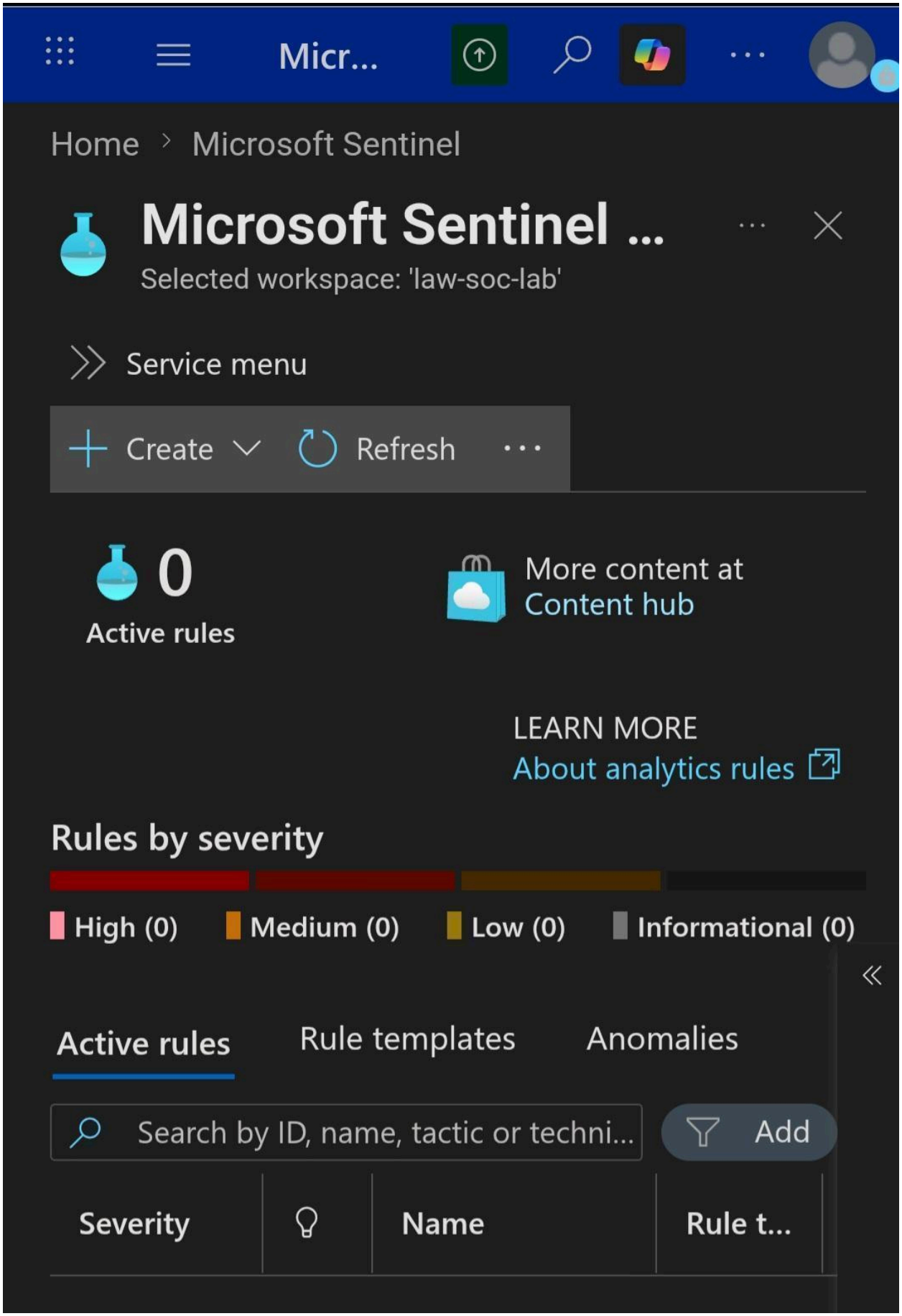
3. Microsoft Sentinel Enablement
Activated Microsoft Sentinel on law-soc-lab workspace

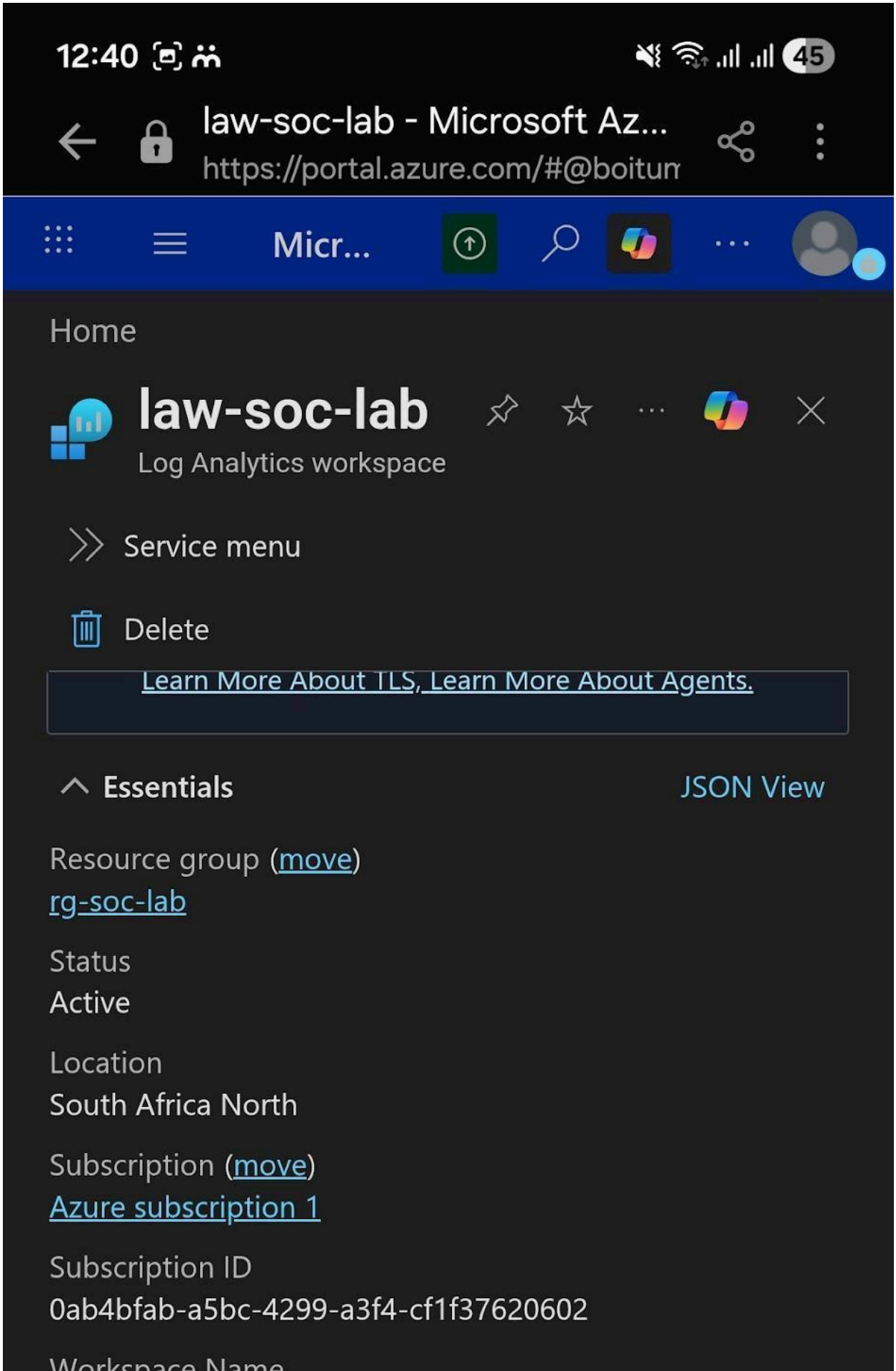
Explored Sentinel dashboard, Incidents, and


Mi...
Upgrade
Home
law-soc-lab
law-soc-lab | ...
Log Analytics workspace
Service menu
Activity
Edit columns
Refresh
Looking for Log Analytics? In Log Analytics you can search for performance, diagnostics, health logs, and more. Visit Log Analytics
Search
Quick Insights
Management Group : None
Subscription : Azure subscription 1
Event severity : All
Timespan : Last 6 hours
Resource group : RG-SOC-LAB
Resource : law-soc-lab
Add Filter
3 items.
Operation name
Status
Time
Ti

Hunting blades
Confirmed role-based access control for SOC operations.
Skills Demonstrated:
Cloud SIEM architecture & deployment
Microsoft Entra ID / Azure AD administration
KQL query development for security logs
MITRE ATT&CK framework implementation
Security data pipeline configuration
Technical documentation

SOC LAB PORTFOLIO-LAW-SOC-LAB


Micr...
Home
Microsoft Sentinel
Microsoft Sentinel ...
Selected workspace: 'law-soc-lab'
Service menu
Create
Refresh
0
Active rules
More content at
Content hub
LEARN MORE
About analytics rules
Rules by severity
High (0)
Medium (0)
Low (0)
Informational (0)
Active rules
Rule templates
Anomalies
Search by ID, name, tactic or techni...
Add
Severity
Name
Rule t...


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law-soc-lab - Microsoft Az...
https://portal.azure.com/#@boitun
Micr...
Home
law-soc-lab
Log Analytics workspace
Service menu
Delete
Learn More About TLS, Learn More About Agents.
Essentials
JSON View
Resource group (move)
rg-soc-lab
Status
Active
Location
South Africa North
Subscription (move)
Azure subscription 1
Subscription ID
0ab4bfab-a5bc-4299-a3f4-cf1f37620602
Workspace Name

Project 2: Brute Force Attack Simulation & Incident Validation

Objective:
Simulate a T1110 Brute Force attack against a cloud identity to validate SIEM detection logic, incident generation, and SOC triage workflow in Microsoft Sentinel.

MITRE ATT&CK Mapping:
Tactic: Credential Access
Technique: T1110 Brute Force
Sub-Technique: T1110.001
Password Guessing
Environment Details:
Platform: Microsoft Azure Sentinel
Workspace: law-soc-lab
Data Source: Microsoft Entra ID Sign

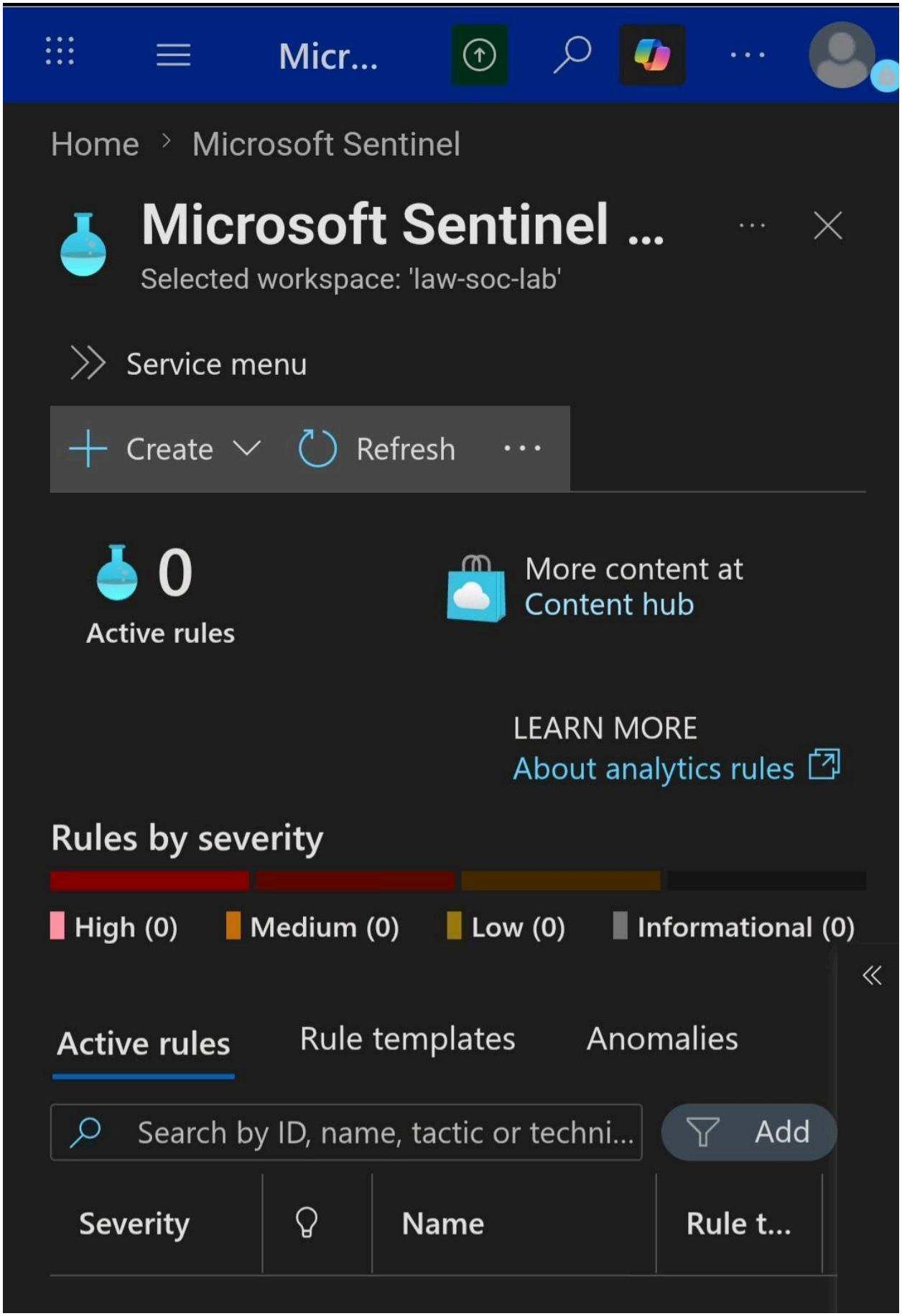
Tasks Completed:
1. Attack Simulation

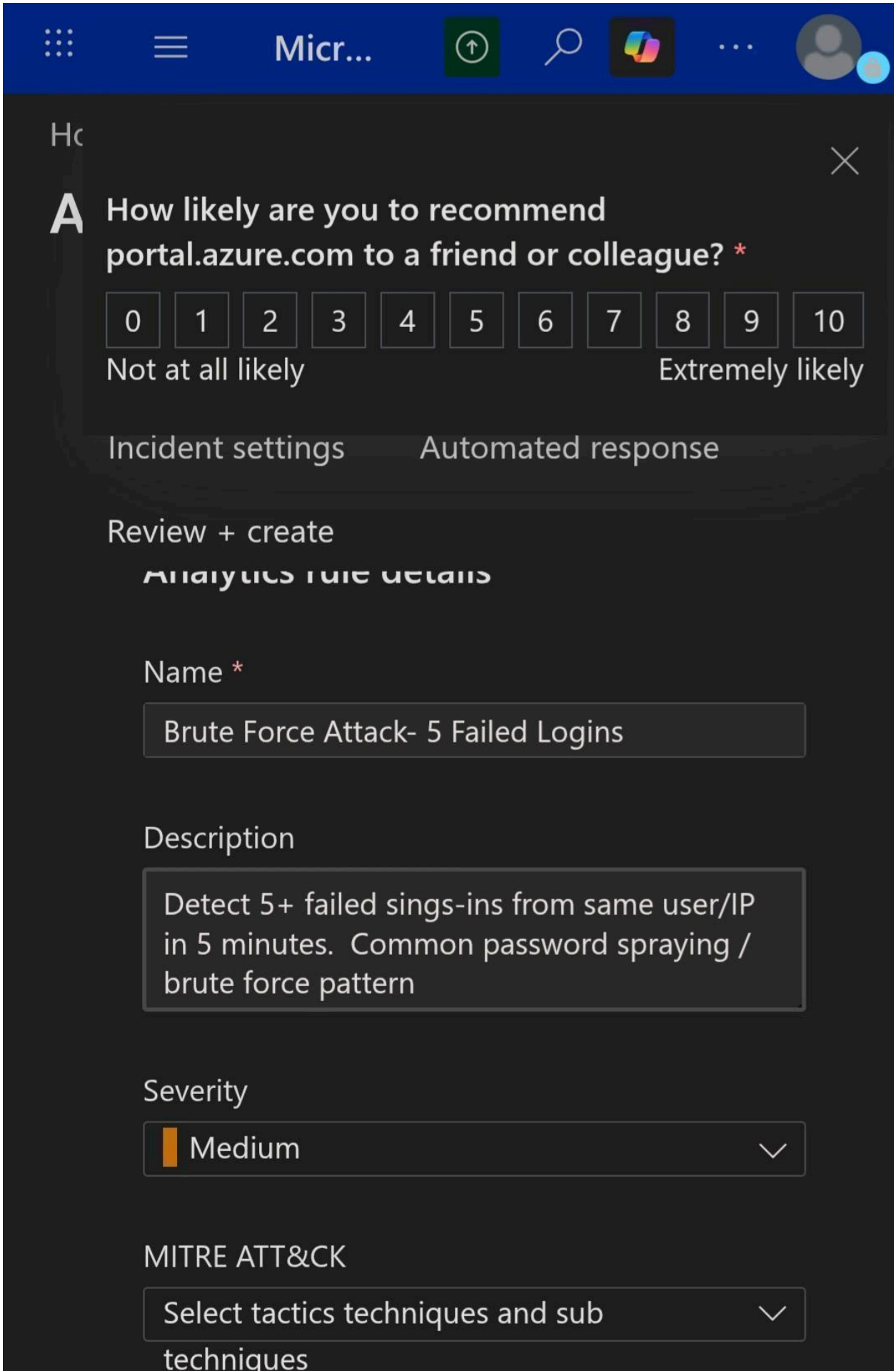
Generated controlled brute force activity against test user boitumeloselowa95
Executed 7 sequential failed authentication attempts from single IP address
Simulated attacker behavior: password guessing via office.com portal
Timestamped attack window for correlation with SIEM ingestion

2. Data Validation & Log Ingestion

Verified Microsoft Entra ID connector

status: Connected (7)
Confirmed real-time streaming of Sign-in logs to Log Analytics Workspace
Acknowledged ingestion delay: 5-15 minutes for cloud log delivery


Micr...
Home
Microsoft Sentinel
Microsoft Sentinel ...
Selected workspace: 'law-soc-lab'
Service menu
Create
Refresh
0
Active rules
More content at
Content hub
LEARN MORE
About analytics rules
Rules by severity
High (0)
Medium (0)
Low (0)
Informational (0)
Active rules
Rule templates
Anomalies
Search by ID, name, tactic or techni...
Add
Severity
Name
Rule t...


Micr...
Hc
A How likely are you to recommend portal.azure.com to a friend or colleague? *
0 1 2 3 4 5 6 7 8 9 10
Not at all likely
Extremely likely
Incident settings
Automated response
Review + create
Name *
Brute Force Attack- 5 Failed Logins
Description
Detect 5+ failed sings-ins from same user/IP in 5 minutes. Common password spraying / brute force pattern
Severity
Medium
MITRE ATT&CK
Select tactics techniques and sub techniques

Project 3 SOC: Account Lockout Investigation

Workflow
Goal

Simulate a real SOC scenario where an attacker tries to brute force an account. You trigger a lockout, then use Sentinel to detect it, investigate, and document the response.

What We Actually Built/Demonstrated

86

1. Attack Simulation

Used boitumeloselowa95@gmail.com to attempt multiple failed logins on login.live.com

Triggered Microsoft's protection: "You've tried to sign in too many times with an incorrect account or password"

This mimics a password spraying or brute force attack you'd see in a real SOC

2. Detection in Sentinel

That failed login activity feeds into your law-soc-lab workspace via the Entra ID/Azure AD connector

Your Brute Force Attack analytics rule from Project 2 is designed to fire on this behavior

Rule set to High severity because repeated failures = potential compromise

3. SOC Investigation Steps

Here's how you'd handle this in a real SOC using your lab:

SOC Phase
What You Do in Sentinel

Triage
Check SigninLogs in Log Analytics for user = boitumeloselowa 95@gmail.com, ResultType != 0

Investigate
Look for: Source IP, location, device, # of attempts, time window

Enrich
Run KQL with geoip to see if IP is from a known bad country/VPN

Respond
Document
Incident would show in Sentinel → options: reset password, block IP, disable account
Add comments to the incident + close with classification: True Positive/Benign Positive



Home > Microsoft Sentinel | Analytics

Analytics rule wizard - ...

Brute force Attack-5 failed logins

General | **Set rule logic** | Incident settings | Automated response | Review + create

Define the logic for your new analytics rule.

Rule query

Any time details set here will be within the scope defined below in the Query scheduling fields.

```
SigninLogs
| where ResultType == "50126"
| summarize FailedCount = count()
| where FailedCount > 5
```

View query results >

Project 4: SOC Role in Blue Team

TryHackMe
Room: SOC Role in Blue Team
Status: Completed 6 tasks, 56 points earned

What This Project Covered:

1. SOC Fundamentals - You learned the day-to-day responsibilities of a SOC Analyst Tier 1
2. People, Process, Technology - The core pillars of a Security Operations Center
3. Security Tools Overview - SIEM, EDR, Firewalls, IDS/IPS and how they fit together
4. Alert Triage Workflow - How to prioritize, investigate, and escalate incidents
5. Blue Team Career Paths - SOC Analyst vs Incident Responder vs Threat Hunter

Key Skills You Proved:

Understanding SOC workflows and escalation paths
Can explain how SIEM fits into the bigger security picture
Knows the difference between reactive monitoring vs proactive threat hunting

09:57 • 86
TryHackMe | Junior Securi...
https://tryhackme.com/room/jrsec

Share your win with your peers

Your progress can inspire others in your community.



Junior Security Analyst Intro completed!

boitumeloselowa95 completed another room on their cyber security journey.

Completed tasks	3
Points earned	32
Streak	2

09:57 • 86
TryHackMe | Junior Securi...
https://tryhackme.com/room/jrsec



Great work boitumeloselowa95!
Room completed!

Your skills are skyrocketing!

Completed tasks	Points earned
3	32

Share your win with your peers

Your progress can inspire others in your community.



SOC Role in Blue Team completed!

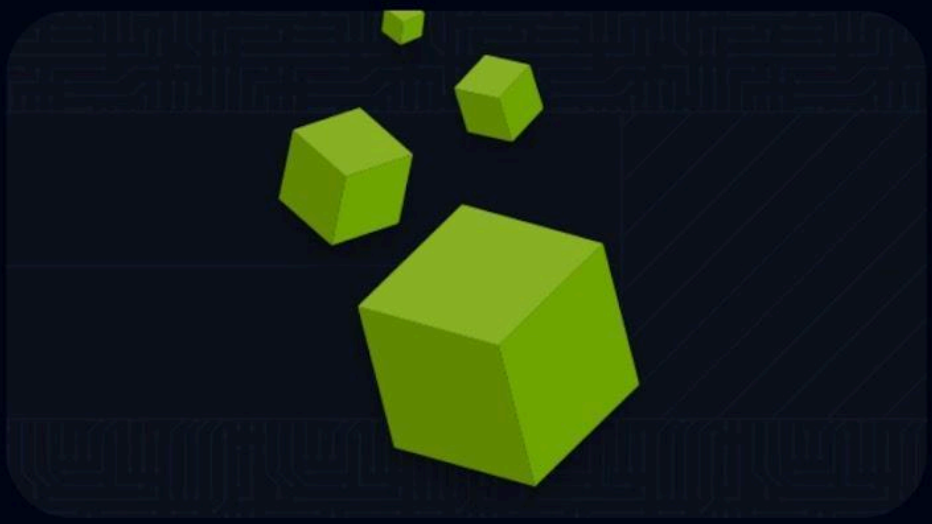
boitumeloselowa95 completed another room on their cyber security journey.

Completed tasks	6
Points earned	56
Streak	2

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Intro to Academy | Hack T...
https://academy.hackthebox.com/a

Close



Congratulations Selowa!

Intro to Academy Module completed!

You earned +10

Share with everyone on:

Project 5: Offensive Security Intro

TryHackMe
Room: Offensive Security Intro
Status: Completed 4 tasks, 32 points earned

Tasks Shown: Think like a Hacker, Starting the Lab, Find Hidden Pages, Attack the Admin Page

What This Project Covered:

1. Attacker Mindset - Learning how hackers think so you can defend better
2. Web App Basics - Hacking a website in a safe legal environment
3. Reconnaissance - Finding hidden pages/directories that devs don't want you to see
4. Accessing Admin Panels - Basic exploitation techniques like directory brute-forcing

Key Skills You Proved:

Hands-on ethical hacking in a controlled lab
Understand common web vulnerabilities: T1190 Exploit Public-Facing Application

Can explain attacks to stakeholders because you've done them yourself
Balances your "Blue Team" defense skills with "Red Team" awareness

Offensive Security Intro

Hack your first website (legally in a safe environment) and experience an ethical hacker's job.

10 min 553,767

Task 1 Think like a Hacker!

Task 2 Starting the Lab

Task 3 Find Hidden Pages

Task 4 Attack the Admin Page